Module 4 Artificial Intelligence Development Framework

Module 4.1: AI Development Framework — 30 Questions

Single Choice Questions (1-10)

Q1 What is the PRIMARY purpose of an AI development framework?

A. To replace hardware completely with software simulation

B. To lower development requirements by providing pre-built models, APIs, and automatic training capabilities

C. To eliminate the need for any programming knowledge

D. To create AI models that only run on mobile devices

Q2 In the model development process, which phase involves "Model building, Model debugging, and Model training"?

A. Inference deployment

B. Training evaluation

C. Model development

D. Data preparation

Q3 PyTorch was developed based on Torch, which was originally implemented in which programming language?

A. Python

B. C++

C. Lua

D. Java

Q4 Which of the following is a KEY characteristic of PyTorch that distinguishes it from static graph frameworks?

A. It uses static computational graphs only

B. It supports dynamic computational graphs that can be built or adjusted during execution

C. It does not support GPU acceleration

D. It requires manual backpropagation implementation

Q5 TensorFlow provides multiple levels of APIs. Which API is specifically designed for quickly building and training models?

A. Distribution Strategy API

B. Keras API

C. TensorFlow Lite API

D. XLA Compiler API

Q6 A company wants to deploy a trained machine learning model on Android smartphones for real-time inference. Which TensorFlow tool is MOST appropriate?

A. TensorFlow.js

B. TensorFlow Extended (TFX)

C. TensorFlow Lite

D. JAX

Q7 JAX combines three key technologies. Which combination is CORRECT?

A. NumPy-like APIs + Autograd + XLA compiler

B. PyTorch tensors + Keras + CUDA

C. Lua scripting + Torch backend + TensorFlow graphs

D. C++ runtime + OpenCL + static graphs only

Q8 The AI framework converts model structure into computational graphs for what primary purpose?
A. To make the model smaller in file size
B. To optimize execution, enable automatic differentiation, and manage memory efficiently
C. To remove the need for training data
D. To convert the model into an image format

Q9 Which statement about distributed training in AI frameworks is CORRECT?
A. It is only supported by PyTorch
B. Some frameworks provide model segmentation and distributed policies for training large models
C. It is not needed for modern deep learning
D. It always reduces model accuracy

Q10 The front-end programming interface of an AI framework allows developers to:
A. Design custom hardware chips
B. Focus on service logic without worrying about underlying mathematical and computing details
C. Eliminate the need for data preprocessing
D. Write code only in assembly language

Multiple Choice Questions (11-16)

Q11 Which of the following are functions of an AI development framework? (Select all that apply)
A. Data pre-processing and conversion
B. Providing development APIs for model building
C. Visualized debugging and tuning capabilities
D. Compiling and executing user code with hardware-software synergy
E. Inference deployment and model delivery

Q12 Which statements about PyTorch are CORRECT? (Select all that apply)
A. It is released by Meta (Facebook)
B. It is a tensor library optimized for deep learning using GPUs and NPUs
C. It uses dynamic computational graphs by default
D. It allows easy debugging by stopping the interpreter to view node outputs
E. It has a rich ecosystem with active community support

Q13 Which statements about TensorFlow and its tools are CORRECT? (Select all that apply)
A. TensorFlow is an end-to-end open-source machine learning platform by Google
B. TensorFlow.js can deploy models on browsers and Node.js
C. TensorFlow Lite is designed for on-device inference on mobile and IoT devices
D. TensorFlow Extended (TFX) is for production ML pipelines
E. The Distribution Strategy API enables distributed training without changing the model definition

Q14 Which statements about JAX are CORRECT? (Select all that apply)
A. It is developed by Google
B. It provides functional programming style
C. It uses XLA-based optimization for high performance
D. It has a larger community ecosystem than PyTorch
E. It supports efficient computing across CPUs, GPUs, and TPUs

Q15 Which activities are part of the AI framework's model development process? (Select all that apply)
A. Data preparation and processing
B. Model building and debugging
C. Model training with iterative optimization
D. Accuracy evaluation
E. Model conversion and inference deployment

Q16 Which capabilities does the AI framework provide for computing optimization and scheduling? (Select all that apply)
A. Converting models into computational graphs
B. Graph optimization for computational efficiency
C. Automatic differentiation based on computational graphs
D. Memory management through graph execution
E. Hardware-based training acceleration

True or False Questions (17-20)

Q17 PyTorch was developed because Torch used the niche programming language Lua, which limited its popularity compared to NumPy-based tools.
A. True
B. False

Q18 TensorFlow 2.X completely removed the Keras API and requires developers to write all models using low-level TensorFlow operations only.
A. True
B. False

Q19 JAX's main disadvantage is its weak community ecosystem compared to frameworks like PyTorch, with fewer documentation resources available.
A. True
B. False

Q20 The AI framework abstracts model structure computing into computational graphs, which enables automatic differentiation but prevents any form of graph optimization.
A. True
B. False

Fill in the Blank Questions (21-24)

Q21 The emergence of deep learning frameworks lowers the __________ for developers. Developers no longer need to compile code starting from complex __________ and __________ algorithms. Instead, they can use existing models to __________ parameters as required, where the model parameters are automatically __________.

Q22 PyTorch is a __________ learning computing framework released by __________. It was developed based on __________, a scientific computing framework supported by many machine learning algorithms. PyTorch is a __________ library optimized for deep learning using __________ and __________. It uses __________ computational graphs by default, allowing programs to dynamically build or adjust graphs during __________.

Q23 TensorFlow is an end-to-end open-source machine learning __________ designed by __________. The main version is TensorFlow __________. It provides multiple levels of APIs, including the advanced __________ API for quickly building models, and the __________ Strategy API for distributed training. For production deployment, __________ Lite is used for mobile/IoT inference, __________.js for browser deployment, and __________ Extended (TFX) for production ML __________.

Q24 JAX is a high-performance __________ computing library developed by __________. It combines __________-like APIs, the automatic differentiation function of __________, and the __________ compiler of TensorFlow. JAX provides __________ programming style and delivers excellent computing performance through __________-based optimization. Its main disadvantage is a relatively __________ community ecosystem compared to __________.

Drag and Drop / Matching Questions (25-30)

Q25 Match each framework to its key characteristic:

Framework	Characteristic
A. PyTorch	1. End-to-end platform by Google with Keras and TFX
B. TensorFlow	2. Dynamic graphs, Python-first, released by Meta
C. JAX	3. Functional programming, XLA optimization, NumPy-like

Write your answer as: A→?, B→?, C→?

Q26 Match each TensorFlow tool to its deployment target:

Tool	Purpose
A. TensorFlow.js	1. Production ML pipelines
B. TensorFlow Lite	2. Browser and Node.js environments
C. TensorFlow Extended (TFX)	3. Mobile and IoT on-device inference

Write your answer as: A→?, B→?, C→?

Q27 Match each model development phase to its activities:

Phase	Activities
A. Preparations	1. Model building, debugging, training
B. Model development	2. Model conversion, inference deployment
C. Inference deployment	3. Data preparation and processing

Write your answer as: A→?, B→?, C→?

Q28 Match each framework to its developing company:

Framework	Company
A. PyTorch	1. Google
B. TensorFlow	2. Meta (Facebook)
C. JAX	3. Google

Write your answer as: A→?, B→?, C→?

Q29 Match each feature to the framework that BEST represents it:

Feature	Framework
A. Dynamic computational graphs by default	1. TensorFlow
B. Functional programming with XLA	2. PyTorch
C. Distribution Strategy API without model changes	3. JAX

Write your answer as: A→?, B→?, C→?

Q30 Match each AI framework component to its function:

Component	**Function**
A. Computational graph	1. Provides easy-to-use programming interfaces
B. Front-end API	2. Enables automatic differentiation and optimization
C. Hardware abstraction	3. Provides hardware access and simplifies deployment

Write your answer as: A→?, B→?, C→?

Scenario-Based Questions (31-35)

Q31 A research team needs to:
Rapidly prototype a new neural network architecture with frequent architecture changes
Debug intermediate layer outputs during training
Deploy on both GPU servers and edge NPU devices
Which framework choice is MOST appropriate?
A. Use PyTorch for prototyping and debugging, then convert for deployment
B. Use only TensorFlow because it is the only framework supporting NPUs
C. Use JAX because it has the largest community and easiest debugging
D. Use raw C++ and CUDA because frameworks limit flexibility

Q32 A tech company has trained a computer vision model and needs to deploy it in three different environments:
A web application running in users' browsers
An Android mobile app for real-time camera inference
A scalable production pipeline with monitoring
Which TensorFlow tools should they use for each environment respectively?
A. TensorFlow.js, TensorFlow Lite, TensorFlow Extended (TFX)
B. TensorFlow Lite for all three environments
C. JAX, PyTorch Mobile, Kubernetes
D. TensorFlow.js for mobile, TFX for browser, Lite for production

Q33 **A lab wants to train a large language model (10B parameters) but has limited GPU memory per device. They need to split the model across multiple GPUs without rewriting the model code.**
Which solution is MOST appropriate?
A. Use TensorFlow's Distribution Strategy API
B. Manually split the model using Python multiprocessing
C. Train on CPU only to avoid memory issues
D. Reduce the model size to fit on one GPU

Q34 A developer is training a model and notices unexpected behavior in a specific layer's output. They need to stop execution mid-training, inspect the tensor values, and modify the computation graph dynamically.
Which framework capability enables this workflow?
A. PyTorch's dynamic computational graphs and debugger integration
B. TensorFlow's static graph compilation
C. JAX's functional programming immutability
D. None of the above; this is impossible in deep learning frameworks

Q35 A startup initially used PyTorch for research due to its flexibility. Now they need to move to production with robust pipeline management, model versioning, and automated deployment.
What is the MOST logical next step?

A. Continue with PyTorch and add custom pipeline tools
B. Migrate to TensorFlow Extended (TFX) for production ML pipelines
C. Rewrite everything in JAX for better performance
D. Abandon deep learning frameworks and use Excel

Advanced Concept Questions (36-40)

Q36 TensorFlow 1.x used static computational graphs, while PyTorch uses dynamic graphs. What is the PRIMARY advantage of dynamic graphs in PyTorch?

A. They execute faster on all hardware
B. They allow dynamic graph modification during execution, enabling easier debugging and flexible architectures
C. They use less memory than static graphs
D. They eliminate the need for backpropagation

Q37 JAX leverages the XLA (Accelerated Linear Algebra) compiler. What is the PRIMARY benefit of XLA?

A. It converts Python code into JavaScript
B. It optimizes and compiles linear algebra operations for efficient execution on accelerators
C. It replaces the need for GPUs entirely
D. It is only used for model visualization

Q38 The AI framework abstracts model structure into computational graphs. What does "automatic differentiation based on computational graphs" mean?

A. The framework manually calculates derivatives using a symbolic math library
B. The framework automatically computes gradients by tracking operations in the graph during forward pass
C. The user must provide derivative formulas for each layer
D. The framework skips gradient calculation to save time

Q39 Modern AI frameworks emphasize "full-stack software-hardware synergy." What does this mean in practice?

A. The framework only runs on custom Huawei chips
B. The framework optimizes computation across software layers down to hardware acceleration (GPUs, NPUs, TPUs)
C. The framework replaces hardware with cloud simulation
D. The framework ignores hardware limitations

Q40 Which statement best describes the evolution trajectory of AI frameworks?

A. From low-level manual implementation → high-level APIs → automated optimization → hardware-software co-design
B. AI frameworks are becoming obsolete and will be replaced by manual coding
C. Future frameworks will only support one type of neural network
D. Frameworks will stop supporting distributed training

Answer Key

Q	Answer	Explanation
1	B	Frameworks lower barriers by providing pre-built models and APIs
2	C	Model development phase includes building, debugging, and training
3	C	Torch was originally in Lua, limiting popularity
4	B	PyTorch's key feature is dynamic computational graphs
5	B	Keras API is for quick model building in TensorFlow
6	C	TensorFlow Lite is designed for mobile/IoT inference
7	A	JAX = NumPy-like + Autograd + XLA
8	B	Computational graphs enable optimization, autodiff, and memory management
9	B	Frameworks like TensorFlow provide distributed training capabilities
10	B	Front-end APIs abstract away low-level math and computing
11	A, B, C, D, E	All are core functions of AI frameworks
12	A, B, C, D, E	All statements about PyTorch are correct
13	A, B, C, D, E	All statements about TensorFlow ecosystem are correct
14	A, B, C, E	D is false; JAX has weaker ecosystem than PyTorch
15	A, B, C, D, E	All are part of the development lifecycle
16	A, B, C, D, E	All are optimization capabilities
17	A - True	PyTorch was created to address Lua's niche status
18	B - False	TensorFlow 2.X integrates Keras, not removes it
19	A - True	JAX's disadvantage is weaker community/docs
20	B - False	Frameworks DO optimize computational graphs
21	requirements, neural networks, backpropagation, configure, trained	Framework abstraction benefits
22	machine, Meta, Torch, tensor, GPUs, NPUs, dynamic, execution	PyTorch characteristics
23	platform, Google, 2.X, Keras, Distribution, TensorFlow, TensorFlow, TensorFlow, pipelines	TensorFlow ecosystem
24	numerical, Google, NumPy, Autograd, XLA, functional, XLA, weak, PyTorch	JAX properties
25	A→2, B→1, C→3	PyTorch=Meta/dynamic, TensorFlow=Google/end-to-end, JAX=functional/XLA
26	A→2, B→3, C→1	js=browser, Lite=mobile, TFX=production
27	A→3, B→1, C→2	Prep=data, Dev=build/train, Deploy=convert/infer
28	A→2, B→1/3, C→1/3	PyTorch=Meta, TensorFlow=Google, JAX=Google
29	A→2, B→3, C→1	Dynamic=PyTorch, Functional/XLA=JAX, Distribution=TensorFlow
30	A→2, B→1, C→3	Graph=autodiff/optimize, API=interfaces, Hardware=access/deploy
31	A	PyTorch for prototyping/debugging, convert for deployment
32	A	js=browser, Lite=mobile, TFX=production pipeline
33	A	Distribution Strategy API handles model parallelism
34	A	PyTorch dynamic graphs + debugger = inspect and modify mid-training
35	B	TFX is designed for production ML pipelines

36	**B**	**Dynamic graphs allow runtime modification and easier debugging**
37	**B**	**XLA optimizes linear algebra for accelerator execution**
38	**B**	**Autodiff tracks operations to automatically compute gradients**
39	**B**	**Full-stack synergy optimizes from software to hardware accelerators**
40	**A**	**Evolution: manual → APIs → auto-optimization → hardware co-design**

Module 4.2: Basics of AI Development Frameworks — 30 Questions

Single Choice Questions (1-10)

Q1 What is the most basic data structure in an AI development framework?
A. Scalar
B. Vector
C. Matrix
D. Tensor

Q2 A zero-order tensor is equivalent to which mathematical concept?
A. Vector
B. Scalar
C. Matrix
D. Array of dimension 3

Q3 A second-order tensor is equivalent to which mathematical concept?
A. Scalar
B. Vector
C. Matrix
D. 4D array

Q4 Based on the standard visualization of tensors, which of the following correctly matches tensor order to its typical representation?
A. 1d-tensor = point, 2d-tensor = line, 3d-tensor = square
B. 1d-tensor = line, 2d-tensor = square, 3d-tensor = cube
C. 1d-tensor = cube, 2d-tensor = square, 3d-tensor = line
D. 1d-tensor = square, 2d-tensor = cube, 3d-tensor = line

Q5 In deep learning frameworks, why is the tensor considered the fundamental data structure rather than matrices or vectors alone?
A. Because tensors are slower and more complex
B. Because tensors can represent data of any dimensionality, unifying scalars, vectors, matrices, and higher-dimensional arrays
C. Because tensors cannot be processed by GPUs
D. Because tensors are only used for image data

Q6 In a typical image classification task using a CNN, the input image is represented as what type of tensor?
A. Zero-order tensor (scalar)
B. First-order tensor (vector)
C. Second-order tensor (matrix)
D. Third-order tensor or higher (e.g., [height, width, channels])

Q7 When two second-order tensors (matrices) are multiplied using matrix multiplication, what is the result?
A. A scalar (zero-order tensor)
B. A vector (first-order tensor)
C. A matrix (second-order tensor)
D. The operation is undefined

Q8 In frameworks like PyTorch and TensorFlow, what is "broadcasting" in the context of tensor operations?

A. Sending tensor data across multiple GPUs

B. Automatically expanding smaller tensors to match the shape of larger tensors for element-wise operations

C. Converting tensors to numpy arrays

D. A technique for reducing tensor dimensions

Q9 How do tensors in AI frameworks like PyTorch differ from NumPy arrays?

A. Tensors cannot perform mathematical operations

B. Tensors support GPU acceleration and automatic differentiation, while NumPy arrays do not

C. Tensors are limited to 2D data only

D. Tensors are slower than NumPy arrays on CPU

Q10 If a tensor has shape (3, 4, 5), what is its order (degree/rank)?

A. 3

B. 4

C. 5

D. 12

Multiple Choice Questions (11-16)

Q11 Which of the following statements about tensors are CORRECT? (Select all that apply)

A. A tensor is a multidimensional array

B. A scalar is a zero-order tensor

C. A vector is a first-order tensor

D. A matrix is a second-order tensor

E. Tensors are limited to a maximum of 3 dimensions

Q12 Which of the following correctly describe tensor orders and their real-world data representations? (Select all that apply)

A. Scalar (0-order): a single number like temperature

B. Vector (1-order): a list of numbers like pixel intensities in a row

C. Matrix (2-order): a grayscale image

D. 3D tensor: a color image with height, width, and RGB channels

E. 4D tensor: a batch of color images [batch_size, height, width, channels]

Q13 Which of the following are valid reasons why tensors are the fundamental data structure in AI frameworks? (Select all that apply)

A. They provide a unified representation for all data types

B. They enable efficient parallel computation on GPUs

C. They support automatic differentiation for gradient computation

D. They can represent complex data structures like batches of sequences

E. They are only used in computer vision tasks

Q14 Which operations are commonly supported by AI frameworks on tensors? (Select all that apply)

A. Element-wise addition and multiplication

B. Matrix multiplication (dot product)

C. Reshaping and transposing

D. Broadcasting for shape compatibility

E. Automatic gradient computation (autograd)

Q15 What advantages do framework tensors have over traditional programming arrays? (Select all that apply)
A. Native GPU acceleration support
B. Built-in automatic differentiation
C. Optimized linear algebra operations
D. Support for distributed computing
E. They are always smaller in memory

Q16 Which of the following correctly identify the number of dimensions for each tensor? (Select all that apply)
A. torch.tensor(5.0) → 0 dimensions
B. torch.tensor([1, 2, 3]) → 1 dimension
C. torch.tensor([[1, 2], [3, 4]]) → 2 dimensions
D. torch.tensor([[[1, 2], [3, 4]], [[5, 6], [7, 8]]]) → 3 dimensions
E. torch.tensor([[[[1]]]]) → 4 dimensions

True or False Questions (17-20)

Q17 A tensor is defined as a multidimensional array that can represent scalars, vectors, matrices, and higher-dimensional data.
A. True
B. False

Q18 In deep learning frameworks, a 4D tensor can represent a batch of color images with dimensions [batch_size, channels, height, width].
A. True
B. False

Q19 Tensors in AI frameworks like PyTorch are fundamentally identical to NumPy arrays and offer no additional capabilities.
A. True
B. False

Q20 The order (or rank) of a tensor refers to the number of dimensions it has, not the total number of elements.
A. True
B. False

Fill in the Blank Questions (21-24)

Q21 A tensor is the most basic __________ in an AI development framework. All data is __________ in tensors. A tensor is defined as a __________ array. A zero-order tensor is a __________, a first-order tensor is a __________, and a second-order tensor is a __________.

Q22 In deep learning, the input to a neural network is typically represented as a tensor. For example, a single grayscale image of size 28×28 pixels is a __________-order tensor with shape __________. A color image of size 224×224 with 3 RGB channels is a __________-order tensor with shape __________. A batch of 32 such color images is a __________-order tensor with shape __________.

Q23 Tensors in AI frameworks support several key capabilities that make them superior to traditional arrays. These include __________ acceleration for parallel computation, __________ __________ for computing gradients automatically during backpropagation, and __________ operations that allow tensors of different but compatible shapes to be combined in element-wise operations.

Q24 The shape of a tensor describes the __________ of elements along each __________. For example, a tensor with shape (2, 3, 4) has __________ dimensions, with __________ elements along the first axis, __________ elements along the second axis, and __________ elements along the third axis. The total number of elements in this tensor is __________.

Drag and Drop / Matching Questions (25-30)

Q25 Match each tensor order to its mathematical equivalent:

Tensor Order	**Mathematical Concept**
A. 0-order	1. Vector
B. 1-order	2. Matrix
C. 2-order	3. Scalar

Write your answer as: A→?, B→?, C→?

Q26 Match each tensor type to a real-world data example:

Tensor Type	**Real-World Example**
A. Scalar (0D)	1. A 224×224 color image
B. Vector (1D)	2. A single temperature reading
C. Matrix (2D)	3. A batch of 10 color images
D. 3D Tensor	4. A grayscale image
E. 4D Tensor	5. A sequence of word embeddings

Write your answer as: A→?, B→?, C→?, D→?, E→?

Q27 Match each tensor operation to its typical result:

Operation	**Result**
A. Adding a scalar to a matrix	1. A vector
B. Matrix multiplication (2D × 2D)	2. A matrix of same shape
C. Summing all elements of a tensor	3. A scalar
D. Reshaping a 2D matrix to 1D	4. A matrix

Write your answer as: A→?, B→?, C→?, D→?

Q28 Match each feature to the tensor capability that enables it:

Feature	**Tensor Capability**
A. Training on GPU	1. Broadcasting
B. Backpropagation	2. GPU tensor support
C. Element-wise ops on different shapes	3. Automatic differentiation
D. Processing batches of data	4. Multi-dimensional structure

Write your answer as: A→?, B→?, C→?, D→?

Q29 Match each tensor property to its correct description:

Property	**Description**
A. Shape	1. The number of dimensions
B. Rank/Order	2. The size along each dimension
C. dtype	3. The total number of elements
D. size/numel	4. The data type (e.g., float32, int64)

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each tensor dimensionality to the type of data it typically represents in deep learning:

Dimensionality	Data Type
A. 1D Tensor	1. Single grayscale image
B. 2D Tensor	2. Batch of sequences
C. 3D Tensor	3. Text sequence (word embeddings)
D. 4D Tensor	4. Batch of color images
E. 5D Tensor	5. Video data (frames, height, width, channels)

Write your answer as: A→?, B→?, C→?, D→?, E→?

Scenario-Based Questions (31-35)

Q31 A data scientist is preparing input for a convolutional neural network. She has 64 color images, each of size 128×128 pixels with 3 color channels (RGB). She needs to process them as a single batch.
What is the shape of the input tensor?

A. (64, 128, 128, 3)
B. (3, 128, 128, 64)
C. (128, 128, 3)
D. (64, 3, 128, 128)

Q32 A researcher has a tensor representing 100 sentences, each with 50 words, and each word represented by a 300-dimensional embedding vector. The current shape is (100, 50, 300). She wants to reshape it to process each sentence independently.
What is the shape after flattening the word and embedding dimensions?

A. (100, 50, 300) — no change needed
B. (100, 15000)
C. (5000, 300)
D. (100, 300, 50)

Q33 A developer has a weight matrix of shape (10, 5) and wants to add a bias vector of shape (5,) to each row. In PyTorch, what happens when he performs this addition?

A. Error: shapes are incompatible
B. The bias vector is automatically broadcasted to shape (10, 5) by replicating it across rows
C. The weight matrix is reduced to shape (5,)
D. The operation requires manual looping over rows

Q34 A model is training on CPU but taking too long. The developer moves all tensors to GPU using .cuda() or .to('cuda'). What is the PRIMARY benefit?

A. The tensors become smaller in memory
B. Parallel computation on thousands of GPU cores speeds up matrix operations
C. The tensors automatically become sparse
D. No benefit; CPUs are always faster for deep learning

Q35 A neural network produces a scalar loss value from a 4D input tensor. During backpropagation, the framework needs to compute gradients for all parameters. Which tensor capability makes this possible?

A. Broadcasting
B. Automatic differentiation (autograd)
C. GPU acceleration
D. Tensor reshaping

Advanced Concept Questions (36-40)

Q36 In frameworks like PyTorch, what is the difference between torch.Tensor and torch.cuda.Tensor?
A. There is no difference; they are identical
B. torch.cuda.Tensor resides in GPU memory and enables GPU computation, while torch.Tensor is in CPU memory
C. torch.Tensor is for scalars only
D. torch.cuda.Tensor cannot perform mathematical operations

Q37 What does "stride" mean in the context of tensor memory layout?
A. The number of elements in the tensor
B. The number of bytes to step in each dimension when traversing the tensor
C. The shape of the tensor
D. The data type of tensor elements

Q38 Why might a tensor operation like .transpose() or .permute() result in a non-contiguous tensor, and what is the implication?
A. The tensor becomes smaller; implication is data loss
B. The tensor shares memory with the original but has different stride layout; some operations require contiguous memory
C. The tensor automatically moves to GPU; implication is faster computation
D. There is no such thing as a non-contiguous tensor

Q39 What is the difference between tensor.view() and tensor.clone() in PyTorch?
A. view() creates a copy of data; clone() creates a view sharing memory
B. view() creates a new view sharing the same underlying data; clone() creates a deep copy with separate memory
C. Both create independent copies with separate memory
D. Both modify the original tensor in-place

Q40 In a computational graph for automatic differentiation, what role do tensors play?
A. Tensors only store final output values
B. Tensors store data values AND track the operations applied to them, enabling gradient flow backward through the graph
C. Tensors are replaced by scalars during backpropagation
D. Tensors are only used for input data, not intermediate computations

Answer Key

Q	Answer	Explanation
1	D	Tensor is the most basic data structure in AI frameworks
2	B	Zero-order tensor = scalar
3	C	Second-order tensor = matrix
4	B	1D=line, 2D=square, 3D=cube
5	B	Tensors unify all dimensionalities
6	D	Color images are 3D+ tensors (H×W×C or more with batch)
7	C	Matrix × Matrix = Matrix
8	B	Broadcasting expands smaller tensors automatically
9	B	Tensors support GPU and autograd; NumPy doesn't natively
10	A	Shape (3,4,5) has 3 dimensions = order 3
11	A, B, C, D	E is false; tensors support unlimited dimensions
12	A, B, C, D, E	All are correct real-world representations
13	A, B, C, D	E is false; tensors are used across all AI domains
14	A, B, C, D, E	All are standard tensor operations
15	A, B, C, D	E is false; tensors aren't always smaller
16	A, B, C, D, E	All correctly identify dimensions
17	A - True	Standard tensor definition
18	A - True	Common format: \[B, C, H, W] or \[B, H, W, C]
19	B - False	Tensors have GPU support and autograd
20	A - True	Order/rank = number of dimensions, not elements
21	data structure, encapsulated, multidimensional, scalar, vector, matrix	Tensor definition and orders
22	2, (28, 28), 3, (224, 224, 3), 4, (32, 224, 224, 3)	Image tensor dimensions
23	GPU, automatic differentiation, broadcasting	Key tensor capabilities
24	number, dimension, 3, 2, 3, 4, 24	Shape properties calculation
25	A→3, B→1, C→2	0D=scalar, 1D=vector, 2D=matrix
26	A→2, B→5, C→4, D→1, E→3	Scalar=temp, Vector=embeddings, Matrix=grayscale, 3D=color image, 4D=batch
27	A→2, B→4, C→3, D→1	Scalar+Matrix=Matrix, MatMul=Matrix, Sum=Scalar, Reshape=Vector
28	A→2, B→3, C→1, D→4	GPU=GPU support, Backprop=autograd, Broadcasting=broadcasting, Batches=multi-dim
29	A→2, B→1, C→4, D→3	Shape=size per dim, Rank=num dims, dtype=data type, size=total elements
30	A→3, B→1, C→2, D→4, E→5	1D=text seq, 2D=grayscale, 3D=batch seq, 4D=batch images, 5D=video
31	A	Standard: (batch, height, width, channels) = (64, 128, 128, 3)
32	B	Flatten last two: 50 × 300 = 15000 → (100, 15000)
33	B	Broadcasting replicates (5,) to (10, 5)
34	B	GPU parallel computation speeds up operations
35	B	Autograd computes gradients automatically
36	B	cuda.Tensor is GPU-resident; Tensor is CPU
37	B	Stride = bytes to step per dimension
38	B	Non-contiguous shares memory with different strides; some ops need contiguous
39	B	view()=shares memory, clone()=deep copy
40	B	Tensors track operations for gradient computation in computational graphs

Module 4.3: PyTorch — 30 Questions

Single Choice Questions (1-10)

Q1 What is the PRIMARY function of the torch core module in PyTorch?
A. To provide only image processing functions
B. To provide tensor operations and linear algebra functions
C. To replace the need for Python entirely
D. To compile models into Java bytecode

Q2 What does the torch.nn module primarily provide in PyTorch?
A. Data loading and preprocessing tools
B. Neural network construction, including layers, loss functions, and activation functions
C. Automatic differentiation engine only
D. Random number generation

Q3 In PyTorch data loading, what is the fundamental difference between Dataset and DataLoader?
A. Dataset loads data in batches; DataLoader defines data format
B. Dataset defines data format and transformations; DataLoader provides batch loading, shuffling, and multi-process loading
C. They are identical and interchangeable
D. Dataset is for training only; DataLoader is for testing only

Q4 When creating a custom dataset by inheriting from torch.utils.data.Dataset, which three methods MUST be overridden?
A. __train__, __test__, __validate__
B. __init__, __len__, __getitem__
C. forward, backward, optimize
D. load, save, transform

Q5 In a PyTorch model class inheriting from nn.Module, which method defines the forward computation (forward propagation)?
A. __backward__
B. forward
C. compute_gradient
D. __init__

Q6 For nn.Conv2d(in_channels, out_channels, kernel_size, stride=1, padding=0), if the input is a color RGB image, what is the value of in_channels?
A. 1
B. 2
C. 3
D. 4

Q7 In torch.nn.RNN, what does setting batch_first=True change?
A. It changes the activation function to ReLU
B. The first dimension of input and output becomes the batch size instead of sequence length
C. It disables the hidden state
D. It makes the RNN bidirectional

Q8 For a multi-class classification problem with 10 classes, which loss function is MOST appropriate?
A. nn.MSELoss()
B. nn.BCELoss()
C. nn.CrossEntropyLoss()
D. nn.L1Loss()

Q9 In torch.optim.SGD, what is the PRIMARY purpose of the momentum parameter?
A. To increase the learning rate over time
B. To accelerate convergence in the relevant direction and suppress oscillation
C. To randomly drop neurons during training
D. To freeze certain layers from updating

Q10 What is the difference between saving model.state_dict() versus saving the entire model in PyTorch?
A. state_dict() saves only parameters (weights/biases); saving the entire model saves architecture + parameters
B. There is no difference between them
C. state_dict() saves the training data
D. Saving the entire model is not possible in PyTorch

Multiple Choice Questions (11-16) ⚠ (Select ALL that apply)

Q11 Which of the following are valid modules in torchvision? (Select all that apply)
A. torchvision.datasets — provides image/video datasets like MNIST, CIFAR, ImageNet
B. torchvision.transforms — supports data augmentation (cropping, rotation, normalization)
C. torchvision.models — provides pre-trained models (ResNet, VGG, AlexNet)
D. torchvision.utils — provides visualization tools (saving tensors as images, image grids)
E. torchvision.optimizers — provides custom optimizers for vision tasks

Q12 Which statements about common PyTorch modules are CORRECT? (Select all that apply)
A. torch.autograd is the automatic differentiation engine for gradient calculation and backpropagation
B. torch.optim provides optimization algorithms (SGD, Adam, AdamW, RMSProp, AdaGrad)
C. torch.device specifies whether tensors and models run on CPU or GPU
D. torch.jit converts dynamic graphs into static ones for optimization and serialization
E. torch.onnx enables model exchange with other deep learning frameworks via ONNX format

Q13 Which statements about PyTorch data loading are CORRECT? (Select all that apply)
A. torchvision.datasets.ImageFolder loads images from folders where each folder represents one class
B. DataLoader provides batch loading, data shuffling, and multi-process data loading
C. transforms.Compose chains multiple preprocessing operations together
D. MNIST dataset can be automatically downloaded by specifying download=True
E. batch_size in DataLoader determines how many samples are processed together in one iteration

Q14 Which of the following are valid activation functions in PyTorch's nn module? (Select all that apply)
A. nn.Sigmoid()
B. nn.Tanh()
C. nn.ReLU()
D. nn.LeakyReLU()
E. nn.Softmax()

Q15 Which statements about PyTorch loss functions are CORRECT? (Select all that apply)
A. nn.L1Loss calculates the average of absolute differences between predicted and actual values
B. nn.MSELoss calculates the average of squared differences
C. nn.CrossEntropyLoss combines LogSoftmax and NLLLoss for multi-class classification
D. nn.BCELoss is used for binary classification tasks
E. nn.CTCLoss is used for automatically aligning unaligned serialized data (e.g., speech recognition)

Q16 Which statements about PyTorch optimizers are CORRECT? (Select all that apply)
A. torch.optim.Adam uses running averages of gradients and gradient squares (betas=(0.9, 0.999))
B. torch.optim.AdamW decouples weight decay from the gradient update, applying it directly to parameters
C. torch.optim.SGD with nesterov=True uses Nesterov accelerated gradient
D. torch.optim.RMSprop uses a smoothing term (alpha) for adaptive learning rates
E. torch.optim.Adagrad adapts learning rates per parameter based on historical gradients

True or False Questions (17-20)

Q17 In PyTorch, when defining a model by inheriting from nn.Module, you must manually define the backward function for backpropagation.
A. True
B. False

Q18 torchvision.datasets.CIFAR10 contains 60,000 32×32 color images divided into 10 categories, with 6,000 images per category.
A. True
B. False

Q19 nn.MaxPool2d(kernel_size=2, stride=2) reduces the spatial dimensions of a feature map by half in both height and width.
A. True
B. False

Q20 In PyTorch training, optimizer.zero_grad() is called before loss.backward() to clear previous gradients, preventing accumulation.
A. True
B. False

Fill in the Blank Questions (21-24)

Q21 PyTorch data is loaded through __________ and __________. The __________ defines the data format and how data is transformed, while the __________ continuously and interactively reads batch data. To create a custom dataset, you inherit from __________ and override the __________, __________, and __________ methods.

Q22torch.nn.Module is the __________ class for constructing models in PyTorch. When defining a model, you must reload the __________ function (for creating model parameters) and the __________ function (for defining forward propagation). PyTorch __________ generates the backward function automatically through __________, so you do not need to define it manually.

Q23
In the LeNet architecture: The first convolutional layer uses __________ 5×5 convolution kernels with stride __________ and padding __________, outputting __________ feature maps of size __________. The pooling layer uses a __________ window with stride __________, reducing dimensions. The second convolutional layer uses __________ 5×5 kernels, outputting __________ feature maps of size __________. After pooling, the feature maps are flattened to __________ and passed through fully connected layers of sizes __________, __________, and finally __________ neurons.

Q24 The standard PyTorch training loop involves: (1) Instantiate the __________, (2) Create a __________ function, (3) Create an __________, (4) Loop over epochs and batches, (5) Call __________ to clear gradients, (6) Forward pass with __________, (7) Calculate __________, (8) Call __________ for backpropagation, (9) Call __________ to update parameters. To save only parameters, use __________; to load them, first instantiate the model then call __________.

Drag and Drop / Matching Questions (25-30)

Q25 Match each PyTorch module to its primary function:

Module	**Function**
A. torch.nn	1. Automatic differentiation engine
B. torch.autograd	2. Tensor operations and linear algebra
C. torch.optim	3. Neural network layers, loss, and activation functions
D. torch	4. Optimization algorithmms for parameter updates

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match each torchvision module to its purpose:

Module	**Purpose**
A. torchvision.datasets	1. Pre-trained models (ResNet, VGG)
B. torchvision.transforms	2. Image/video datasets (MNIST, CIFAR, ImageNet)
C. torchvision.models	3. Visualization utilities (image grids, saving)
D. torchvision.utils	4. Data augmentation and preprocessing

Write your answer as: A→?, B→?, C→?, D→?

Q27 Match each layer specification to its position in LeNet:

Layer Spec	**LeNet Position**
A. 6 feature maps, 28×28	1. After second pooling
B. 16 feature maps, 10×10	2. First convolutional layer output
C. 120 neurons	3. Second convolutional layer output
D. 5×5×16 flattened	4. First fully connected layer input

Write your answer as: A→?, B→?, C→?, D→?

Q28 Match each optimizer to its distinctive parameter:

Optimizer	**Key Parameter**
A. SGD	1. betas=(0.9, 0.999)
B. Adam	2. alpha=0.99 (smoothing term)
C. RMSprop	3. nesterov=True option
D. AdamW	4. Decoupled weight decay

Write your answer as: A→?, B→?, C→?, D→?

Q29 Match each loss function to its appropriate use case:

Loss Function	Use Case
A. nn.CrossEntropyLoss	1. Binary classification
B. nn.BCELoss	2. Regression with absolute error
C. nn.MSELoss	3. Multi-class classification
D. nn.L1Loss	4. Regression with squared error

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each code snippet to the training step it performs:

Code Snippet	Training Step
A. optimizer.zero_grad()	1. Update model parameters
B. loss.backward()	2. Clear accumulated gradients
C. optimizer.step()	3. Compute gradients via backpropagation
D. outputs = net(inputs)	4. Forward propagation

Write your answer as: A→?, B→?, C→?, D→?

Scenario-Based Questions (31-35)

Q31 A developer is building a CNN for 10-class image classification. The input images are 32×32 RGB. She designs the following architecture:
Conv layer: 16 filters, 3×3 kernel, padding=1
ReLU activation
MaxPool: 2×2, stride=2
Conv layer: 32 filters, 3×3 kernel, padding=1
ReLU activation
MaxPool: 2×2, stride=2
Flatten
Fully connected: 32×8×8 → 128
Fully connected: 128 → 10
What is the shape of the tensor after the second MaxPool layer?

A. (batch_size, 16, 16, 16)
B. (batch_size, 32, 8, 8)
C. (batch_size, 32, 16, 16)
D. (batch_size, 16, 8, 8)

Q32 A researcher has a folder structure:
plain
```
dataset/
  cats/
    cat1.jpg, cat2.jpg, ...
  dogs/
    dog1.jpg, dog2.jpg, ...
```
She wants to load this for training with batch size 32, shuffled, and 4 worker processes.
Which code correctly accomplishes this?

A. torchvision.datasets.MNIST(root='./dataset', train=True)
B. torchvision.datasets.ImageFolder(root='./dataset', transform=transform); DataLoader(dataset, batch_size=32, shuffle=True, num_workers=4)
C. torch.utils.data.Dataset(root='./dataset', batch_size=32)
D. torchvision.datasets.CIFAR10(root='./dataset', download=True)

Q33 A team wants to use a pre-trained ResNet-50 for a new 5-class classification task. They need to:
Load the pre-trained model
Replace the final fully connected layer
Freeze all layers except the final one
Train on their dataset
Which approach is MOST correct?
A. Load torchvision.models.resnet50(pretrained=True), replace model.fc with nn.Linear(2048, 5), set requires_grad=False for all parameters except model.fc
B. Train ResNet-50 from scratch on the 5-class data
C. Load the model and train all layers without modification
D. Use torch.jit to convert the model to static graph first

Q34 A developer notices that during training, the loss decreases but model performance on validation does not improve. She suspects vanishing gradients in early layers.
Which PyTorch capability can help diagnose this?
A. torch.autograd to inspect gradient magnitudes at each layer during loss.backward()
B. torch.optim.SGD with higher learning rate
C. torchvision.transforms to augment data
D. torch.jit to compile the model

Q35 A model is trained on GPU but needs to be deployed on a CPU-only server for inference. The saved model contains CUDA tensors.
What is the CORRECT loading procedure?
A. torch.load('model.pth', map_location='cpu') then model.eval()
B. Directly load on CPU; PyTorch automatically converts
C. Load on GPU first, then move to CPU with .cpu()
D. The model cannot be loaded on CPU if saved from GPU

Advanced Concept Questions (36-40)

Q36 PyTorch uses dynamic computational graphs by default. What is the implication of this for the forward method?
A. The graph is built once and reused for all inputs
B. A new computational graph is built on-the-fly for each forward pass, enabling dynamic architectures
C. The graph must be defined in a separate configuration file
D. Dynamic graphs are slower and should always be avoided

Q37 In torch.autograd, what does requires_grad=True indicate?
A. The tensor will be stored on GPU
B. The tensor requires gradient computation, and operations on it will be tracked for backward pass
C. The tensor is immutable
D. The tensor will be normalized automatically

Q38 Where should nn.BatchNorm2d typically be placed in a CNN architecture relative to activation functions?
A. After the activation function
B. Before the activation function (between convolution and activation)
C. Only at the input layer
D. Batch normalization is never used with CNNs

Q39 A training process shows loss plateauing after epoch 50. Which PyTorch component should be used to automatically reduce the learning rate when loss stops improving?

A. torch.optim.lr_scheduler.ReduceLROnPlateau

B. torch.nn.Dropout

C. torch.utils.data.DataLoader

D. torchvision.transforms.RandomCrop

Q40 What is the critical difference between model.train() and model.eval() in PyTorch?

A. model.train() enables gradient computation; model.eval() disables it

B. model.train() sets layers like Dropout and BatchNorm to training mode; model.eval() sets them to evaluation mode (e.g., Dropout disabled, BatchNorm uses running stats)

C. model.train() runs on GPU; model.eval() runs on CPU

D. There is no functional difference; they are for code readability only

Answer Key

Q	Answer	Explanation
1	B	`torch` provides tensor operations and linear algebra
2	B	`torch.nn` provides layers, loss functions, and activations
3	B	Dataset defines format/transforms; DataLoader handles batching/shuffling
4	B	Must override `__init__`, `__len__`, `__getitem__`
5	B	`forward` defines forward propagation
6	C	RGB images have 3 channels
7	B	`batch_first=True` makes first dimension = batch size
8	C	CrossEntropyLoss for multi-class classification
9	B	Momentum accelerates convergence, suppresses oscillation
10	A	`state_dict()` = parameters only; full model = architecture + parameters
11	A, B, C, D	E is false; torchvision has no optimizers module
12	A, B, C, D, E	All statements are correct
13	A, B, C, D, E	All statements about data loading are correct
14	A, B, C, D, E	All are valid activation functions in PyTorch
15	A, B, C, D, E	All descriptions of loss functions are correct
16	A, B, C, D, E	All optimizer descriptions are correct
17	B - False	PyTorch auto-generates backward via autograd
18	A - True	CIFAR10: 60,000 images, 10 classes, 6,000 per class
19	A - True	MaxPool 2×2 stride 2 halves spatial dimensions
20	A - True	`zero_grad()` clears gradients before new backward pass
21	Dataset, DataLoader, Dataset, DataLoader, torch.utils.data.Dataset, **init**, **len**, **getitem**	Data loading architecture
22	base, **init**, forward, automatically, torch.autograd	Model construction in PyTorch
23	six, 1, 2, 6, 28×28, 2×2, 2, sixteen, 16, 10×10, 16×5×5=400, 120, 84, 10	LeNet architecture details
24	model, loss, optimizer, optimizer.zero_grad(), model(inputs), loss, loss.backward(), optimizer.step(), torch.save(model.state_dict(), ...), model.load_state_dict(torch.load(...))	Training loop and save/load
25	A→3, B→1, C→4, D→2	nn=layers, autograd=diff, optim=update, torch=tensor ops
26	A→2, B→4, C→1, D→3	datasets=data, transforms=augment, models=pretrained, utils=visualize
27	A→2, B→3, C→4, D→1	6\@28×28=conv1, 16\@10×10=conv2, 120=fc1 input, 5×5×16=flatten after pool2
28	A→3, B→1, C→2, D→4	SGD=nesterov, Adam=betas, RMSprop=alpha, AdamW=decoupled decay

29	A→3, B→1, C→4, D→2	CrossEntropy=multi-class, BCE=binary, MSE=squared regression, L1=absolute regression
30	A→2, B→3, C→1, D→4	zero_grad=clear, backward=gradients, step=update, net(inputs)=forward
31	B	32\@8×8 after second pool: 32 filters, 32/2/2=8 spatial
32	B	ImageFolder for folder-per-class structure + DataLoader with proper params
33	A	Standard transfer learning: load pretrained, replace FC, freeze early layers
34	A	autograd allows inspecting gradient magnitudes to diagnose vanishing gradients
35	A	`map_location='cpu'` loads CUDA tensors directly to CPU
36	B	Dynamic graphs are built per forward pass, enabling flexible architectures
37	B	`requires_grad=True` tracks operations for gradient computation
38	B	BatchNorm is placed between Conv and Activation (before activation)
39	A	`ReduceLROnPlateau` reduces LR when metric stops improving
40	B	`train()` enables Dropout/BN training mode; `eval()` disables Dropout, uses running BN stats

Module 4.4: AI Application Development Process

Single Choice Questions (1-10)

Q1 According to the AI Application Development Process diagram, which is the FIRST step after requirement analysis?

A. Data preparation
B. Network building
C. Environment setup
D. Model training

Q2 In the environment setup phase of AI application development, which of the following is NOT typically included?

A. Framework installation
B. Cloud environment configuration
C. Model deployment to production servers
D. Python environment setup

Q3 The flower image dataset used in the ResNet-50 example contains how many flower types?

A. 3
B. 4
C. 5
D. 10

Q4 Who proposed ResNet-50 and in which year did it win first place in ILSVRC image classification?

A. Geoffrey Hinton, 2012
B. Yann LeCun, 2014
C. Kaiming He from Microsoft Labs, 2015
D. Yann LeCun, 2015

Q5 What is the key innovation of ResNet that enables deeper networks without degradation?

A. Dropout layers
B. Residual connections (skip connections) that add the input x to F(x)
C. Batch normalization only
D. Using only fully connected layers

Q6 When training a model for flower classification using ResNet-50, which approach is described as fine-tuning the last-layer parameters of a pre-trained model?

A. Training from scratch on the flower dataset
B. Using a pre-trained model on ImageNet and fine-tuning for 5-class flower classification
C. Randomly initializing all weights
D. Using only the first layer of ResNet-50

Q7 Based on the model training and inference diagram, what is the KEY difference between training and inference?

A. Training uses more data than inference
B. Training requires updating model weights using loss and correct results; inference only produces prediction results without weight updates
C. Inference requires GPU; training uses CPU
D. Training is faster than inference

Q8 Which of the following is NOT a primary motivation for model compression?

A. Reducing inference latency
B. Decreasing deployment costs (storage, compute, VRAM)
C. Increasing the number of model parameters
D. Enabling deployment on resource-constrained devices like mobile phones

Q9 In model pruning, what happens to less important weights in the weight matrix?

A. They are increased in value
B. They are changed to zero elements, sparsifying the network
C. They are moved to a different layer
D. They are converted to floating-point infinity

Q10 In knowledge distillation, the large complex network is referred to as the ________, and the smaller network is referred to as the ________.

A. student, teacher
B. teacher, student
C. master, apprentice
D. server, client

Multiple Choice Questions (11-16)

Q11 Which of the following are phases in the AI Application Development Process? (Select all that apply)

A. Requirement analysis
B. Environment setup
C. Data preparation
D. Network building
E. Model training
F. Application deployment

Q12 Which activities are included in data preprocessing for the flower classification application? (Select all that apply)

A. Reading the dataset
B. Generating data augmentation operations (cropping, rotation, flipping, normalization)
C. Processing the generated dataset
D. Cleaning and sorting collected service data (if not using open-source datasets)
E. Automatically training the model without any preprocessing

Q13 Which statements about ResNet-50 are CORRECT? (Select all that apply)

A. It was proposed by Kaiming He from Microsoft Labs
B. It won first place in ILSVRC 2015 image classification
C. It uses residual connections to mitigate degradation in deep networks
D. It is only suitable for natural language processing tasks
E. It enables deeper network structures than previous architectures

Q14 Which approaches are described for training the flower classification model? (Select all that apply)

A. Start training from scratch based on the dataset
B. Fine-tune a pre-trained model on ImageNet, adjusting last-layer parameters for 5-class flower classification
C. Use random weights without any pre-training
D. Adjust hyper-parameter combinations during training
E. Use only unsupervised learning methods

Q15 Which model compression methods are discussed in the module? (Select all that apply)

A. Model pruning
B. Knowledge distillation
C. Quantization to 8-bit integers
D. Removing all activation functions
E. Converting models to ONNX format

Q16 Which statements about model inference and deployment are CORRECT? (Select all that apply)

A. Inference does not require updating model weight parameters
B. torch.no_grad() should be used during inference to disable gradient calculation
C. The model should be set to eval() mode during inference
D. Model inference has lower hardware requirements than training
E. Models can be deployed on dedicated inference servers or optimized for mobile devices

True or False Questions (17-20)

Q17 The AI Application Development Process starts with environment setup, followed by requirement analysis.

A. True
B. False

Q18 Cloud environments like ModelArts provide more computing power and storage than local environments, making them suitable for large-scale model training.

A. True
B. False

Q19 In knowledge distillation, the goal is to make the small network (student) achieve similar performance to the large network (teacher) while requiring fewer computing resources.

A. True
B. False

Q20 Model pruning always improves model performance while reducing model size.

A. True
B. False

Fill in the Blank Questions (21-24)

Q21

The AI Application Development Process consists of six main phases: __________ analysis, __________ setup, __________ preparation, __________ building, __________ training, and __________ deployment.

Q22 ResNet-50 was proposed by __________ from __________ in __________ and won first place in the __________ 2015 image classification competition. The key innovation is the __________ network, which uses __________ connections to add the input __________ to the output __________, effectively mitigating __________ and enabling __________ network structures.

Q23 Model compression is essential because increasing model parameters impacts __________ performance (increased data transfer overhead when memory bandwidth is limited, and CPU-only deployment when VRAM is insufficient) and __________ cost (more storage, computing resources, and devices required). Two compression methods are __________, which removes less important weights and sparsifies the network, and __________, which transfers knowledge from a large __________ network to a smaller __________ network.

Q24 During model inference, gradient calculation should be disabled using __________, and the model should be switched to __________ mode. The model weight file is loaded by first creating the model's __________, then loading the __________. For deployment, models are usually placed on dedicated __________ servers, and after optimization (such as __________), they can also be deployed on __________ phones and other devices.

Drag and Drop / Matching Questions (25-30)

Q25 Match each development phase to its activities:

Phase	Activities
A. Environment setup	1. Hyper-parameter definition, model structure building
B. Network building	2. Framework installation, cloud environment use
C. Model training	3. Model training, model saving, model testing
D. Application deployment	4. Model deployment, model prediction

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match each compression method to its description:

Method	Description
A. Model pruning	1. Transfer knowledge from large network to small network
B. Knowledge distillation	2. Remove less important weights, sparse the network
C. Model quantization	3. Reduce precision of weights (not explicitly covered)

Write your answer as: A→?, B→?, C→?

Q27 Match each dataset characteristic to its value:

Characteristic	Value
A. Total images	1. 633
B. Daisies	2. 898
C. Dandelions	3. 3670
D. Roses	4. 641

Write your answer as: A→?, B→?, C→?, D→?

Q28 Match each training approach to its description:

Approach	Description
A. Training from scratch	1. Use pre-trained ImageNet model, adjust last layer for new classes
B. Fine-tuning	2. Initialize random weights, train entirely on new dataset
C. Transfer learning	3. Leverage knowledge from pre-trained model for new task

Write your answer as: A→?, B→?, C→?

Q29 Match each ResNet component to its function:

Component	Function
A. Weight layer	1. Add input directly to output
B. ReLU activation	2. Learn feature transformations
C. Identity skip connection	3. Introduce non-linearity
D. F(x) + x	4. Residual learning formula

Write your answer as: A→?, B→?, C→?, D→?

Q30 Match each aspect to its impact of large models:

Aspect	Impact
A. Limited memory bandwidth	1. More storage and devices needed
B. Insufficient VRAM	2. Increased data transfer overhead
C. Larger weight size	3. Model must deploy on CPU, lowering inference speed

Write your answer as: A→?, B→?, C→?

Scenario-Based Questions (31-35)

Q31 A team is building an AI application to classify medical images. They have completed requirement analysis and installed PyTorch. Their next steps are: collect 10,000 labeled images, design a CNN architecture, train the model, and deploy it to hospital servers.

Which sequence correctly orders the remaining phases?

A. Data preparation → Network building → Model training → Application deployment
B. Network building → Data preparation → Model training → Application deployment
C. Application deployment → Data preparation → Network building → Model training
D. Model training → Data preparation → Network building → Application deployment

Q32 A startup has limited training data (2,000 images) for a 10-class product classification task. They want to achieve high accuracy quickly without massive compute resources.

Which strategy is MOST appropriate?

A. Train ResNet-50 from scratch on the 2,000 images
B. Use a pre-trained ResNet-50 on ImageNet and fine-tune the last layer for 10 classes
C. Build a custom architecture with 100 layers from scratch
D. Use only data augmentation without any pre-trained model

Q33 A company trained a large transformer model for chatbot inference. The model has 10 billion parameters and requires 40GB VRAM. They need to deploy it on consumer GPUs with 8GB VRAM.

Which compression approach combination is MOST suitable?
A. Only use knowledge distillation to create a smaller student model
B. Apply model pruning to reduce parameters, then knowledge distillation to maintain performance
C. Increase the batch size to fit the model
D. Buy more expensive GPUs and do not compress

Q34 A developer deployed a trained model for real-time image classification. Users report slow response times. The model is currently running in training mode with gradient calculation enabled.

What changes should be made to optimize inference?
A. Switch to model.eval() and use torch.no_grad() context
B. Increase the learning rate
C. Add more dropout layers
D. Train the model for more epochs

Q35 A research team needs to train a ResNet-50 on 1 million images. Their local machines have 8GB RAM and no GPU. They need results within one week.

What is the MOST appropriate solution?
A. Train on local CPU; it will finish eventually
B. Use a cloud environment like ModelArts with GPU instances
C. Reduce the dataset to 1,000 images to fit local resources
D. Use a smaller model like LeNet-5 regardless of accuracy requirements

Advanced Concept Questions (36-40)

Q36 In the residual block F(x) + x, why is learning the residual F(x) easier than learning the direct mapping H(x)?

A. Because F(x) is always larger than x
B. If the optimal mapping is close to identity, learning small perturbations (residuals) is easier than learning the full mapping from scratch
C. Because ReLU activation makes gradients always positive
D. Because skip connections eliminate the need for backpropagation

Q37 When fine-tuning a pre-trained ResNet-50 for flower classification (5 classes) from ImageNet (1000 classes), which layers are typically modified and why?

A. All layers are retrained from scratch
B. Only the final fully connected layer is replaced (1000 → 5 outputs) and fine-tuned, while earlier layers remain frozen or use very low learning rates because they learn general features
C. Only the first convolutional layer is modified
D. All batch normalization layers are removed

Q38 Model pruning removes less important weights to create sparse networks. What is the critical challenge in pruning?

A. Pruning always improves both speed and accuracy
B. Finding the balance between model size reduction and performance deterioration
C. Pruning is impossible on convolutional layers
D. Pruned models cannot run on GPUs

Q39 In knowledge distillation, what does the "soft target" from the teacher network provide that one-hot hard labels do not?

A. The soft target provides class probabilities that encode inter-class similarities and richer information about the teacher's confidence
B. The soft target is just a random vector
C. The soft target has no advantage over hard labels
D. The soft target only works for binary classification

Q40 Which statement BEST describes the relationship between the AI Application Development Process phases?

A. Each phase is completely independent with no feedback loops
B. The process is iterative; insights from later phases (e.g., model training results) may require returning to earlier phases (e.g., data preparation or network building)
C. The process is linear and never requires revisiting previous phases
D. Only the deployment phase matters; other phases are optional

Answer Key

Q	Answer	Explanation
1	C	After requirement analysis comes environment setup
2	C	Model deployment is the final phase, not environment setup
3	C	5 types: daisies, dandelions, roses, sunflowers, tulips
4	C	Kaiming He, Microsoft Labs, 2015, ILSVRC winner
5	B	Residual connections: F(x) + x, the key innovation
6	B	Fine-tuning pre-trained model on ImageNet for 5-class flowers
7	B	Training updates weights via loss; inference only predicts
8	C	Increasing parameters is the opposite of compression motivation
9	B	Pruning changes less important weights to zero
10	B	Teacher = large network, Student = small network
11	A, B, C, D, E, F	All six are phases in the development process
12	A, B, C, D	E is false; preprocessing is always required
13	A, B, C, E	D is false; ResNet is for computer vision
14	A, B, D	C and E are not described approaches
15	A, B	Quantization and ONNX are not discussed as compression methods
16	A, B, C, D, E	All are correct inference/deployment practices
17	B - False	Process starts with requirement analysis, then environment setup
18	A - True	Cloud provides more compute and storage
19	A - True	Student learns from teacher with fewer resources
20	B - False	Pruning may cause performance deterioration; needs balance
21	requirement, environment, data, network, model, application	Six phases of AI development
22	Kaiming He, Microsoft Labs, 2015, ILSVRC, Residual, skip/identity, x, F(x), degradation, deeper	ResNet-50 history and innovation
23	inference, deployment, pruning, knowledge distillation, teacher, student	Model compression motivations and methods
24	torch.no_grad(), eval, instance, state_dict, inference, compression, mobile	Inference and deployment procedures
25	A→2, B→1, C→3, D→4	Environment=setup, Network=building, Training=train/save/test, Deploy=deploy/predict
26	A→2, B→1, C→3	Pruning=sparse weights, Distillation=transfer knowledge, Quantization=reduce precision
27	A→3, B→1, C→2, D→4	Total=3670, Daisies=633, Dandelions=898, Roses=641
28	A→2, B→1, C→3	Scratch=random init, Fine-tune=pre-trained + adjust last layer, Transfer=leverage pre-trained
29	A→2, B→3, C→1, D→4	Weight layer=learn features, ReLU=non-linearity, Skip=add input, F(x)+x=residual formula
30	A→2, B→3, C→1	Bandwidth=transfer overhead, VRAM=CPU deployment, Weight=more storage/devices
31	A	Correct sequence: Data → Network → Train → Deploy
32	B	Fine-tuning pre-trained model is best for limited data
33	B	Pruning + distillation combination for large model compression
34	A	eval() + no_grad() are standard inference optimizations

35	B	Cloud GPU environment is necessary for large-scale training
36	B	Learning residuals near identity is easier than full mapping
37	B	Replace final FC layer; freeze early layers with general features
38	B	Critical challenge: balance between size reduction and performance
39	A	Soft targets encode class similarities and teacher confidence
40	B	Process is iterative with feedback loops between phases